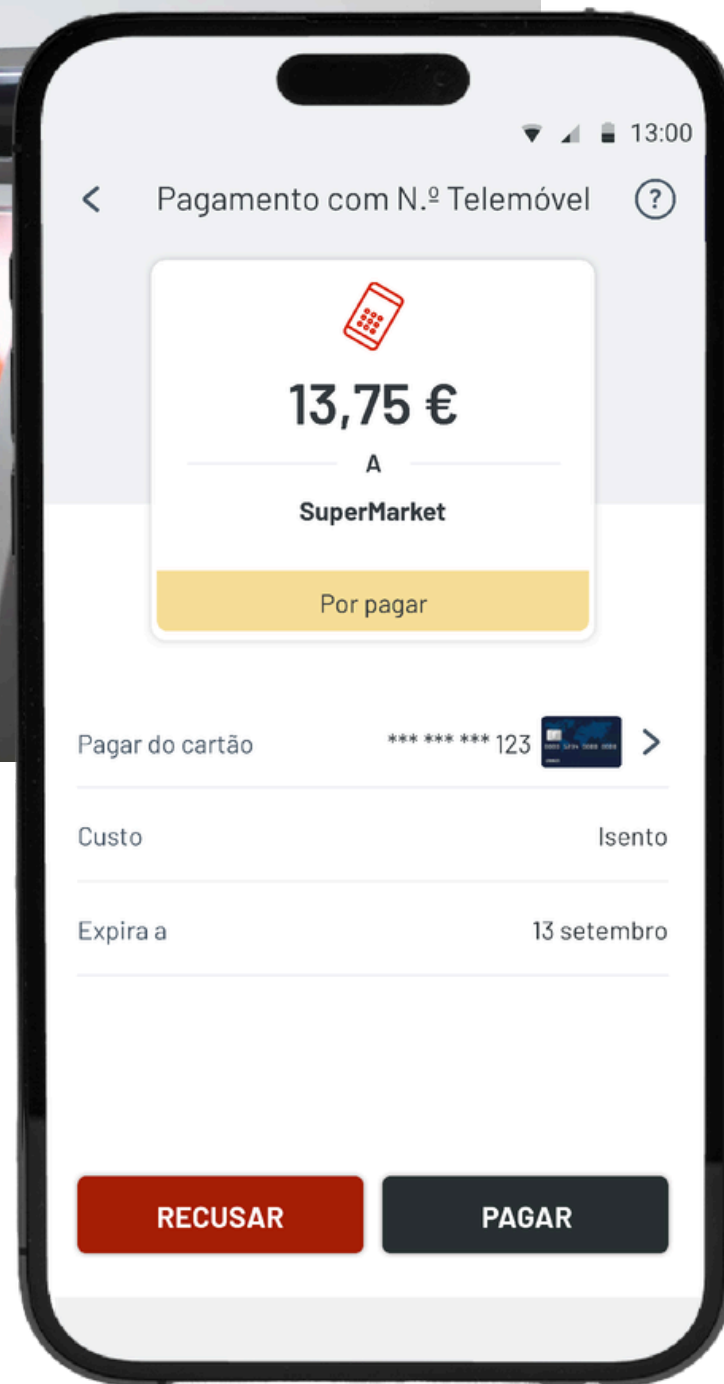


Model-View-Controller

Padrões para a Camada de Apresentação





Bank of America

Regular Checking - 0873

Sign Out

Available Balance

\$7,900.00

Search Transactions

Amount Included in Available Balance

Processing	- \$45.00
Online Banking Transfer CONF# 39TRSMTN; SMITH	\$7,900.00
Processing	- \$55.00
Online Banking Transfer CONF# 39TRVPYAN; DOE	\$7,945.00
03/25/2016	- \$250.00
SIMMONS TRUCK 03/25 #000505591 PURCHASE SIMMONS TRUCK T BRACEY VA CKCD 5542 4635730003908594	\$8,000.00
03/18/2016	\$100.00
BKOFAMERICA MOBILE 03/18 3417580533 DEPOSIT *MOBILE NC	\$8,250.00
03/04/2016	- \$50.00
CITY OF CHARLOTE-SERVICES NC Bill Payment	\$8,150.00
03/03/2016	- \$30.00
TRANSFER TO ACCT #3456 ON 03/03 VIA CALL AGENT	\$8,100.00
02/20/2016	\$8,087.62
BKOFAMERICA MOBILE 02/20 45678766 DEPOSIT *MOBILE NC	\$8,130.00
02/19/2016	\$20.00
Agent Assisted transfer from CHK 8767 Confirmation# 2185390024	\$42.38
02/19/2016	\$20.00
Agent Assisted transfer from CHK 8767 Confirmation# 1485416077	\$22.38
02/18/2016	\$0.85
Online scheduled transfer from CHK 8741 Confirmation# 1011849519	\$2.38
02/10/2016	\$1.03
BKOFAMERICA MOBILE 02/10 2356789876 DEPOSIT *MOBILE NC	\$1.53
02/08/2016	- \$0.50
Agent Assisted transfer to CHK 8767 Confirmation# 3196014802	\$0.50

Account number

Show Account Number

Routing Numbers

Paper and Electronic

053000196

Use this routing number to order checks, set up direct deposits and outgoing payments to other financial institutions.

Wires

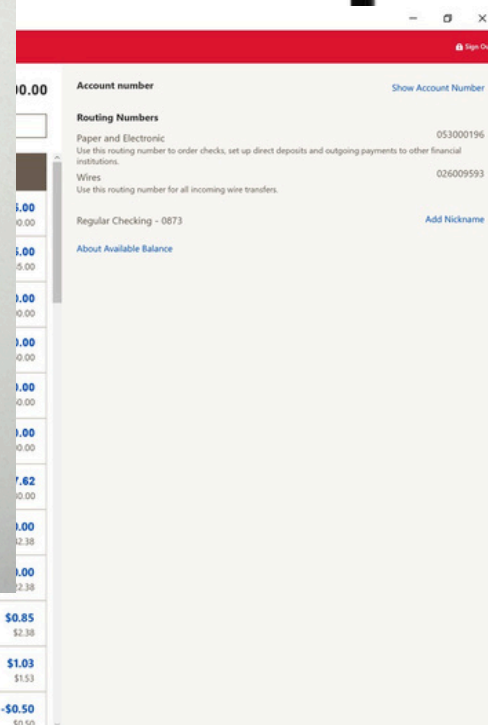
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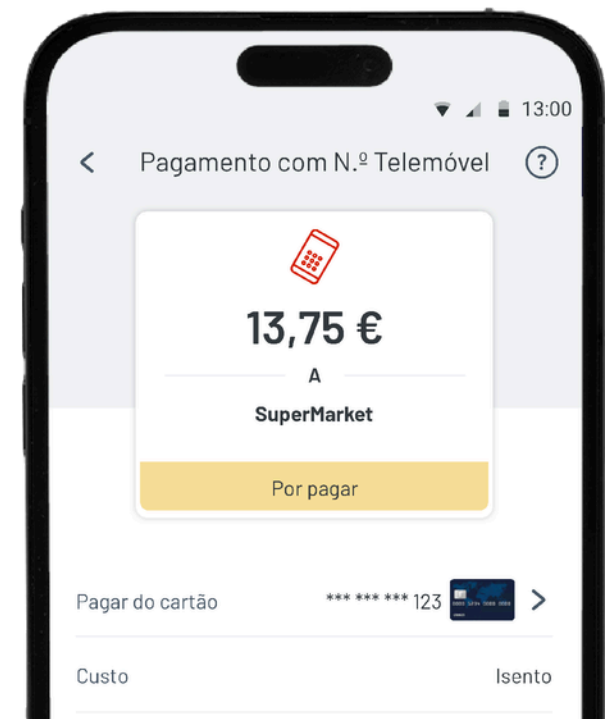
Use this routing number for all incoming wire transfers.

Regular Checking - 0873

Add Nickname

About Available Balance





Bank of America

Regular Checking - 0873

Available Balance **\$7,900.00**

Search Transactions

Amount Included in Available Balance	
Processing	-\$45.00
Online Banking Transfer CONF# 39TRSMTN; SMITH	\$7,900.00
Processing	-\$55.00
Online Banking Transfer CONF# 39TRVPYAN; DOE	\$7,945.00
03/25/2016	-\$250.00
SIMMONS TRUCK 03/25 #000505591 PURCHASE SIMMONS TRUCK T BRACEY VA CKCD 5542 463573000390859	\$8,000.00
03/18/2016	\$100.00
BKOFAMERICA MOBILE 03/18 3417580533 DEPOSIT *MOBILE NC	\$8,250.00
03/04/2016	-\$50.00
CITY OF CHARLOTTE-SERVICES NC Bill Payment	\$8,150.00
03/03/2016	-\$30.00
TRANSFER TO ACCT #3456 ON 03/03 VIA CALL AGENT	\$8,100.00
02/20/2016	\$8,087.62
BKOFAMERICA MOBILE 02/20 45678766 DEPOSIT *MOBILE NC	\$8,130.00
02/19/2016	\$20.00
Agent Assisted transfer from CHK 8767 Confirmation# 2185390024	\$42.38
02/19/2016	\$20.00
Agent Assisted transfer from CHK 8767 Confirmation# 1485416077	\$22.38
02/18/2016	\$0.85
Online scheduled transfer from CHK 8741 Confirmation# 1015849519	\$2.38
02/10/2016	\$1.03
BKOFAMERICA MOBILE 02/10 2356789876 DEPOSIT *MOBILE NC	\$1.53
02/08/2016	-\$0.50
Agent Assisted transfer to CHK 8767 Confirmation# 31960154802	\$0.50

Account number [Show Account Number](#)

Routing Numbers

Paper and Electronic 053000196
Use this routing number to order checks, set up direct deposits and outgoing payments to other financial institutions.

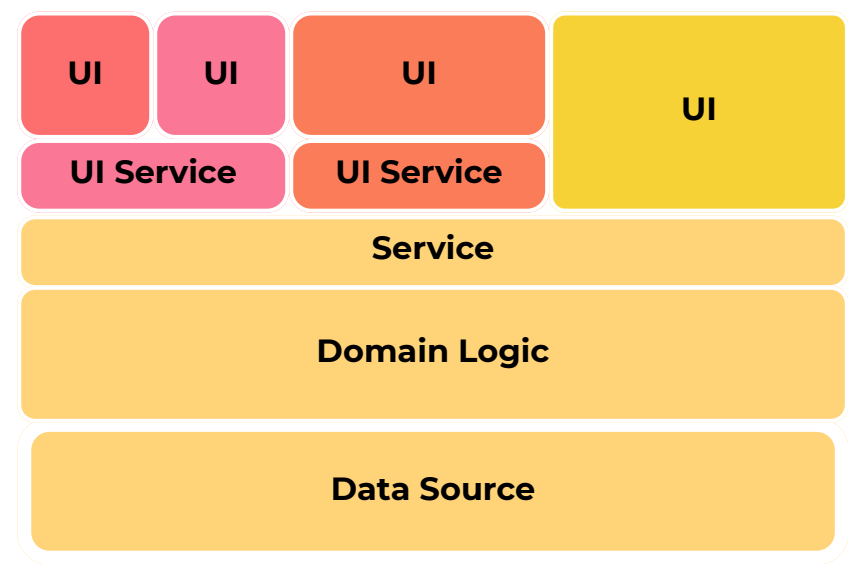
Wires 026009593
Use this routing number for all incoming wire transfers.

Regular Checking - 0873 [Add Nickname](#)

[About Available Balance](#)

Padrões para as várias camadas

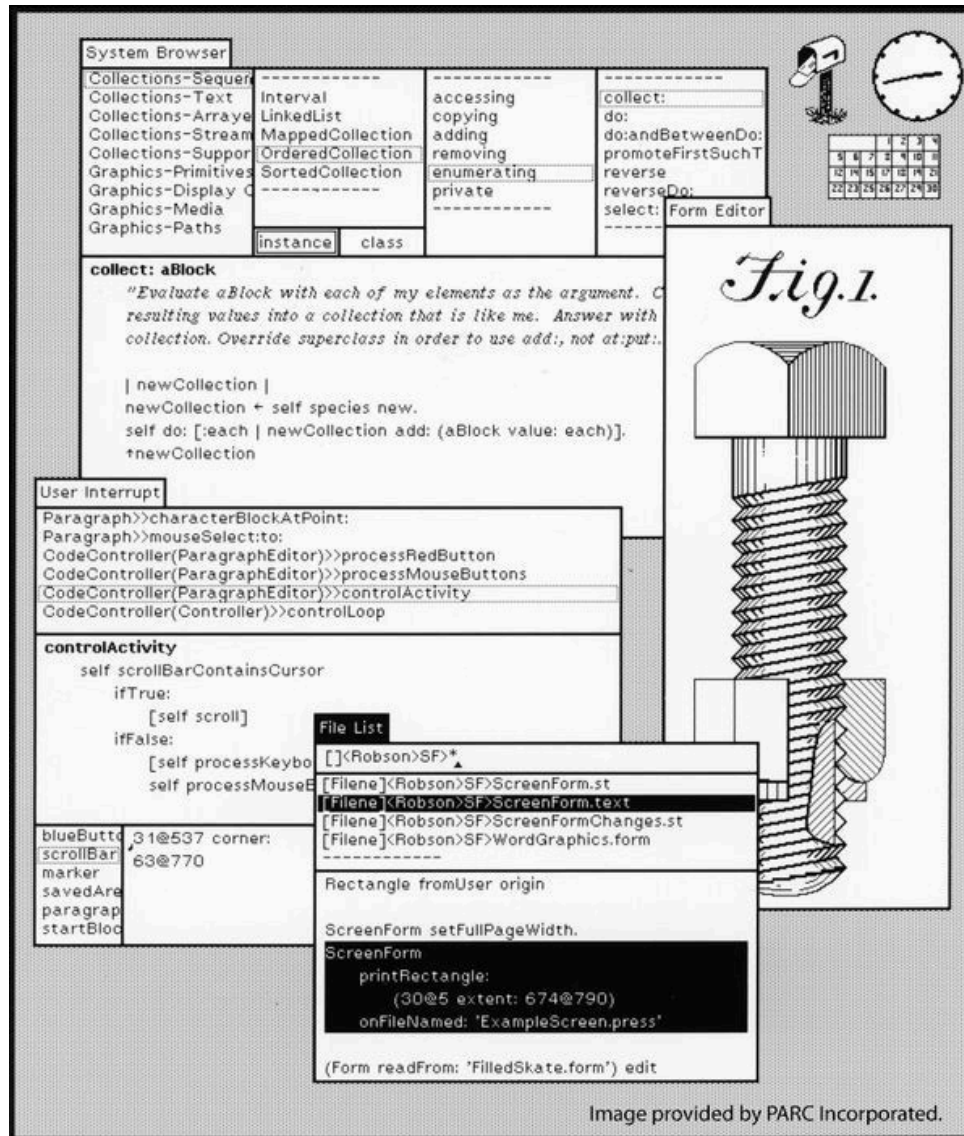
Camada de Apresentação	MVC Front Controller Page Controller Page template
Camada de Negócio	Service Layer
	Domain Model
	Table Module
	Transaction Script
Camada de Dados	Row data gateway
	Table data gateway
	Active Record
	Data Mapper



- A camada de apresentação é responsável pela interação com **humanos** ou **aplicações cliente**
- Podem existir várias UIs em simultâneo.
- A UI pode expor diferentes protocolos, conforme os clientes.

MVC Clássico

SmallTalk-80



Xerox PARC has been at the heart of numerous revolutionary computer developments, including [laser printing](#), [Ethernet](#), the modern [personal computer](#), GUI ([graphical user interface](#)) and [desktop paradigm](#), [object-oriented programming](#), [ubiquitous computing](#), [electronic paper](#), a-Si ([amorphous silicon](#)) applications, the [computer mouse](#), and VLSI ([very-large-scale integration](#)) for [semiconductors](#).^{[6][5]}

MVC Clássico

SmallTalk-80

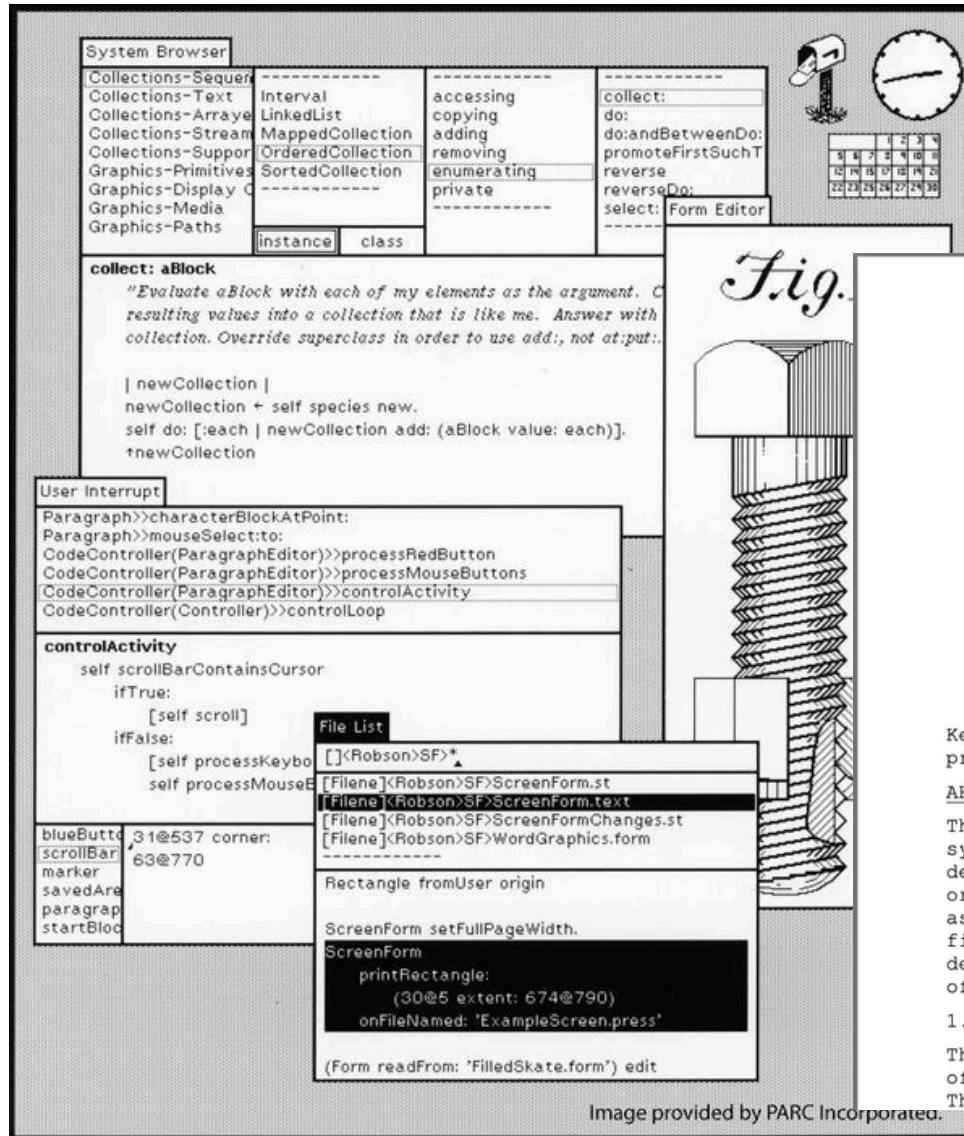


Image provided by PARC Incorporated.

Xerox PARC has been at the heart of numerous revolutionary computer developments, including [laser printing](#), [Ethernet](#), the modern [personal computer](#), GUI ([graphical user interface](#)) and [desktop paradigm](#), [object-oriented programming](#), [ubiquitous computing](#), [electronic paper](#), a-Si ([amorphous silicon](#)) applications, the [computer mouse](#), and VLSI ([very-large-scale integration](#)) for

Fig.

ADMINISTRATIVE CONTROL IN THE SHIPYARD

Preprint of paper to be delivered at the ICCAS Conference. Tokyo, August 1973.

Professor Trygve Reenskaug
Central Institute for Industrial Research
Blindern, Oslo 3, Norway

(Scanned 2003-2-26 by Trygve Reenskaug. Some typos fixed, some probably added)

Key words: Production planning, production monitoring, shipbuilding, data processing, organisation, decentralisation.

ABSTRACT

The organisation of a yard may be strongly influenced by the information system it takes into use. The converse should also be true: Systems development should be subordinated the desired evolution of the organisation. The principle of Communicating Data Processes is introduced as a tool for building information systems that are easily moulded for fitting an existing organisation and for adaption to its continuous development. A description of the principle is given, as well as an outline of a new system designed for testing it out in practical shipbuilding.

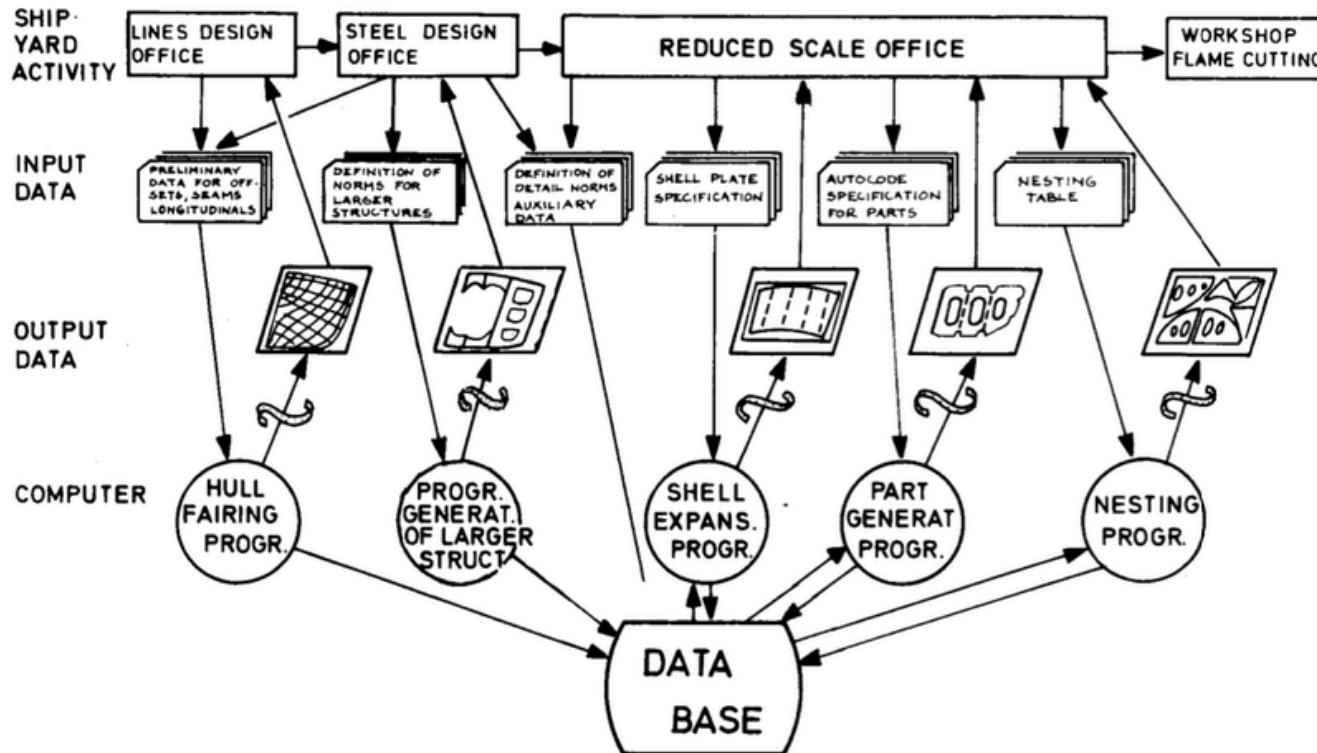
1. INTRODUCTION

The modern ship is an extremely complex product requiring the co-ordination of a great variety of skills, materials and machines for its production. The good management of a shipyard therefore presupposes advanced tools for

MVC Clássico

Sn

comprehensive (2). In time, the yard will be utilising a number of such systems, and the need for some sort of integration is growing. Today, functions are manually linked together in the various departments, each department being responsible for parts of several functions. Integration of each function is by formal and informal communication between departments.



ADMINISTRATIVE CONTROL IN THE SHIPYARDPage

Xerox PARC has been at the heart of numerous revolutionary computer developments, including [laser printing](#),

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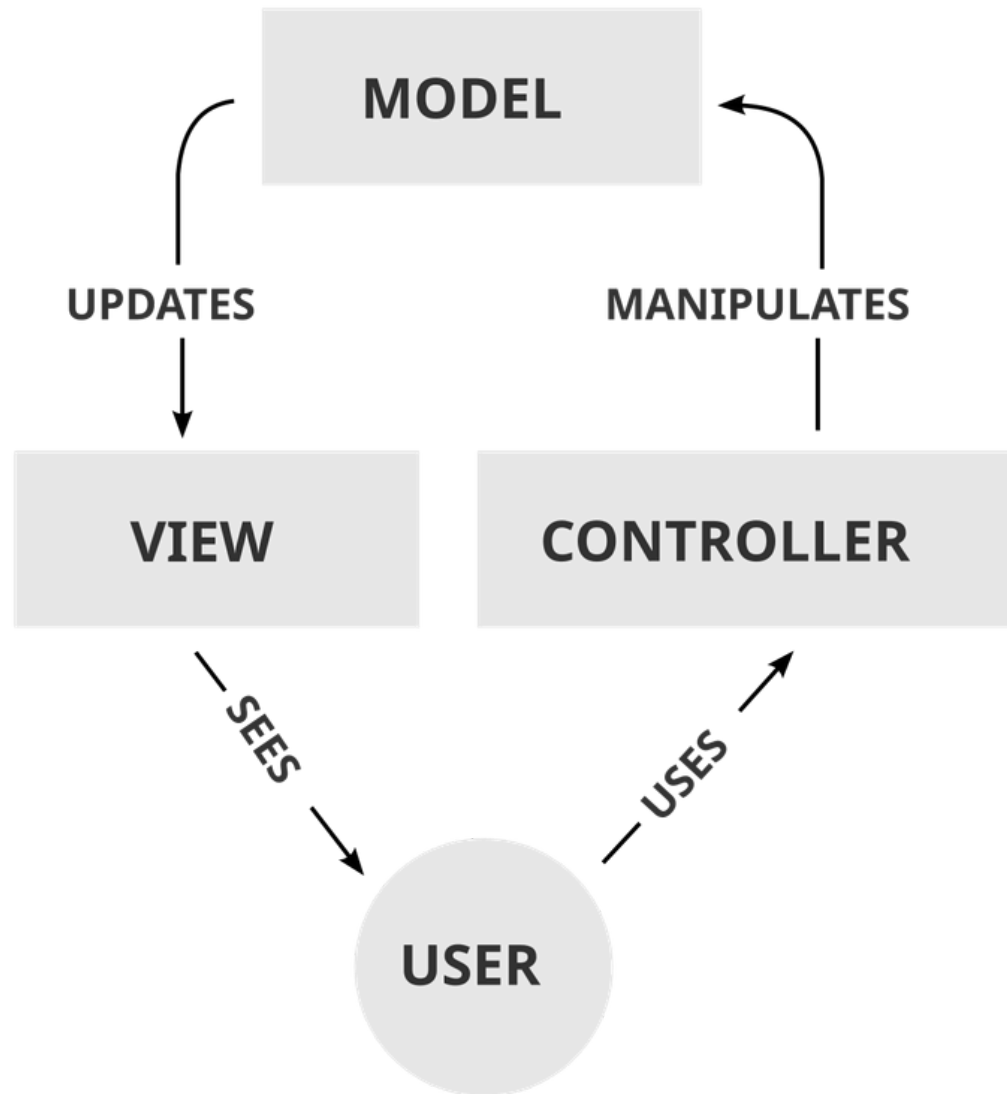
tools for

Model, View e Controller

A perspectiva clássica

In his "Thing-Model-View-Editor" paper (see Reenskaug, Trygve, "Thing-Model-View-Editor"), Dr. Reenskaug defines the following terms:

1. **Thing:** "Something that is of interest to the user." (A real-world concept.)
2. **Model:** "An active representation of an abstraction in the form of data in a computing system." (Abstraction of the Thing.)
3. **View:** "One or more pictorial representations of the model."
4. **Editor:** "An interface between a user and one or more Views." (Later called "Controller" and "Tool.")



O MVC e a Web

- Perl, PHP e ASP (clássico) misturavam HTML com código (com SQL, CSS e pequenas linhas de javascript).
- Várias frameworks vieram tentar implementar o MVC na web:
 - Struts (Model=bean, Controller=Servlet, View=~HTML)
 - Rails (Model=Entity, Controller=~Servlet, View=~HTML)
 - Django (Model=Entity, View=~Servlet, Template=~HTML)
 - .NET (Model=Entity, View=UI, Presenter=Controller)

O MVC e a Web

- Perl, PHP e ASP (clássico) misturavam HTML com código (com S
- Veb:



Os vários MVCs

E os diferentes níveis de abstracção

- Interface web (burra) que precisa de lógica no servidor para **qualquer** interacção

vs

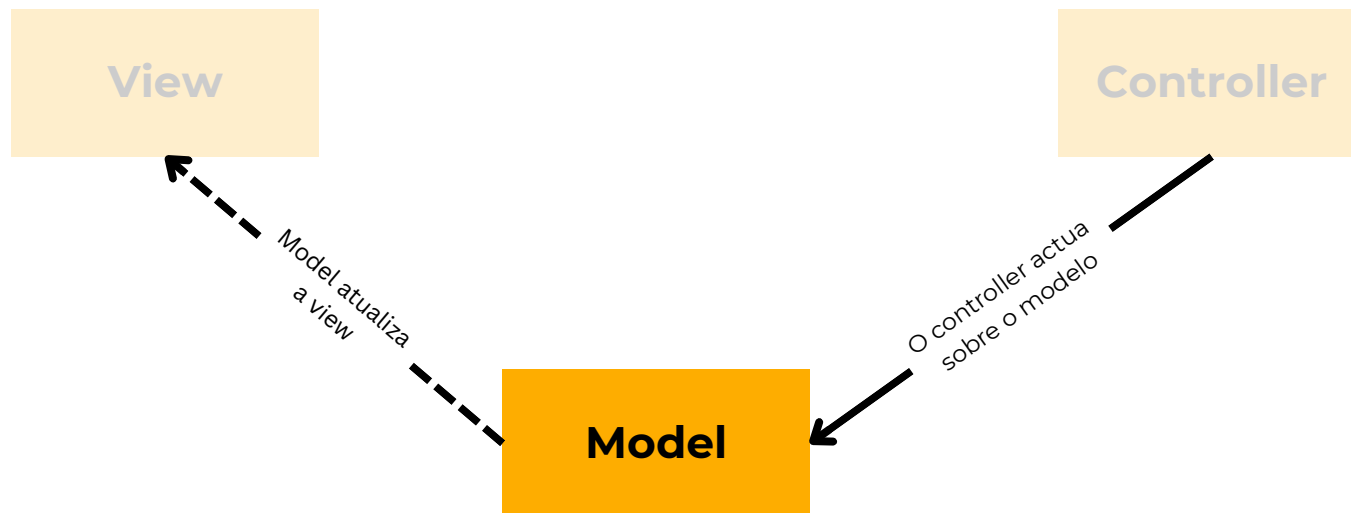
- Interface nativa, com lógica imediata

- Um componente é um botão

vs

- Um componente é um formulário

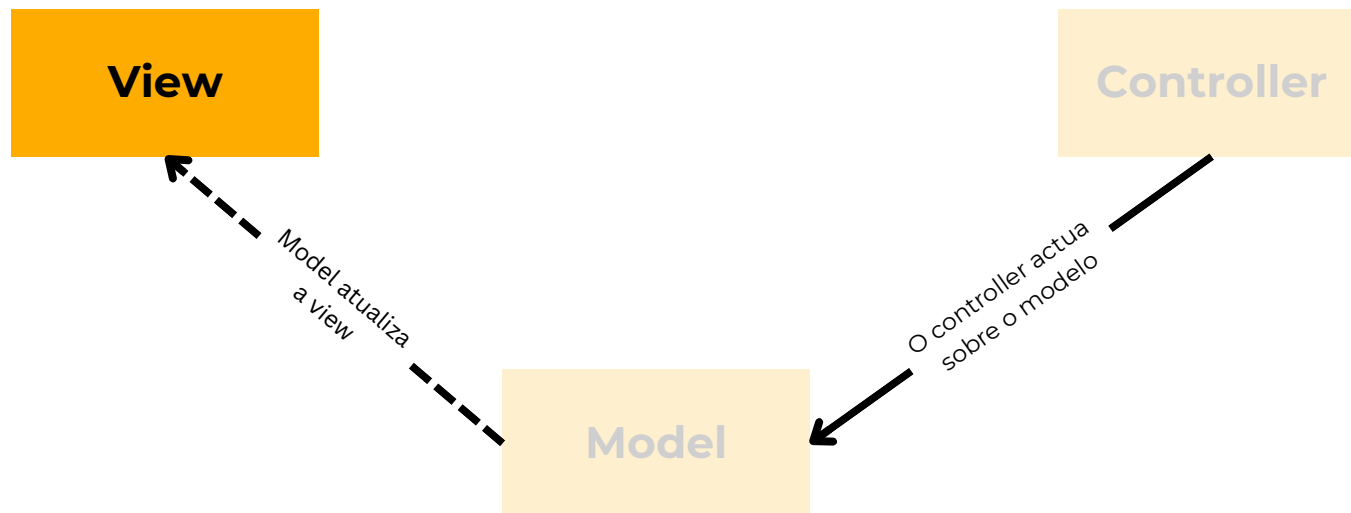
Model View Controller



- **Model**

- Representa os dados no MVC. Implementa a lógica de negócio e o processamento dos dados (garante estado e integridade).
- É responsável por responder a pedidos e comandos do Controlador.
- Notifica a Vista sobre mudanças

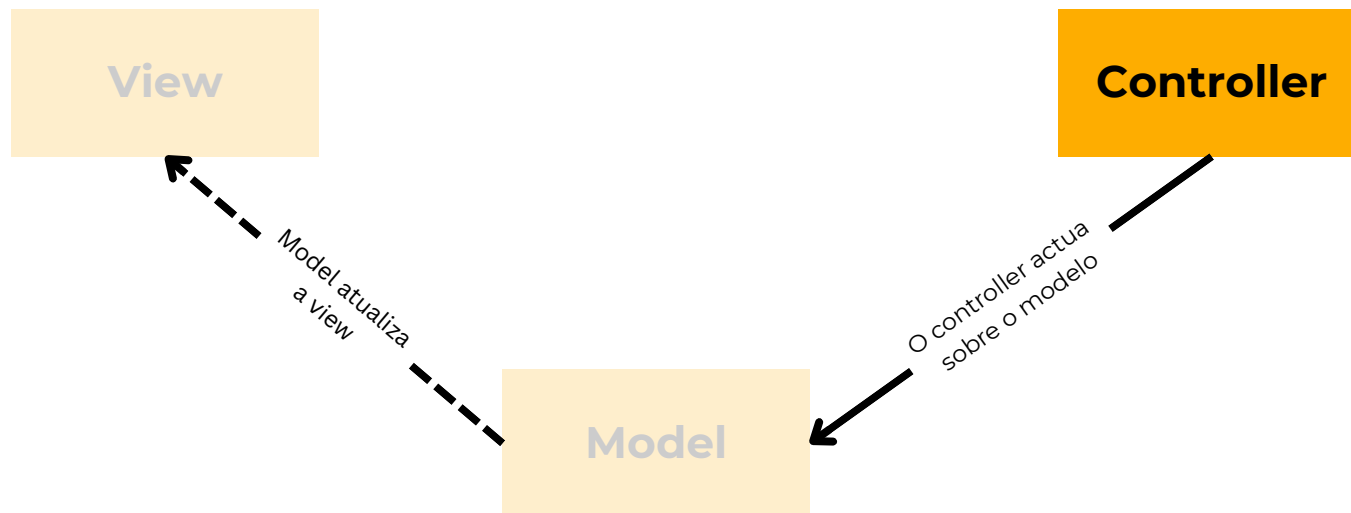
Model View Controller



- **View**

- Apresenta os dados ao utilizador
- Mostra a informação com base no Modelo.
- Encaminha interações do utilizador para o Controlador
- Separado da regra de negócio

Model View Controller



- **Controller**

- Recebe e interpreta comandos da Vista
- Atualiza o Modelo.
- Não deve conter manipulação dos dados ou lógica de apresentação
- Acesso concorrente

Model View Controller

Define a estrutura e aparência dos dados que são mostrados.



Define como o input vai alterar a aplicação. (Ex: clique de ratos, abertura de ficheiros, etc...)



Model atualiza a view

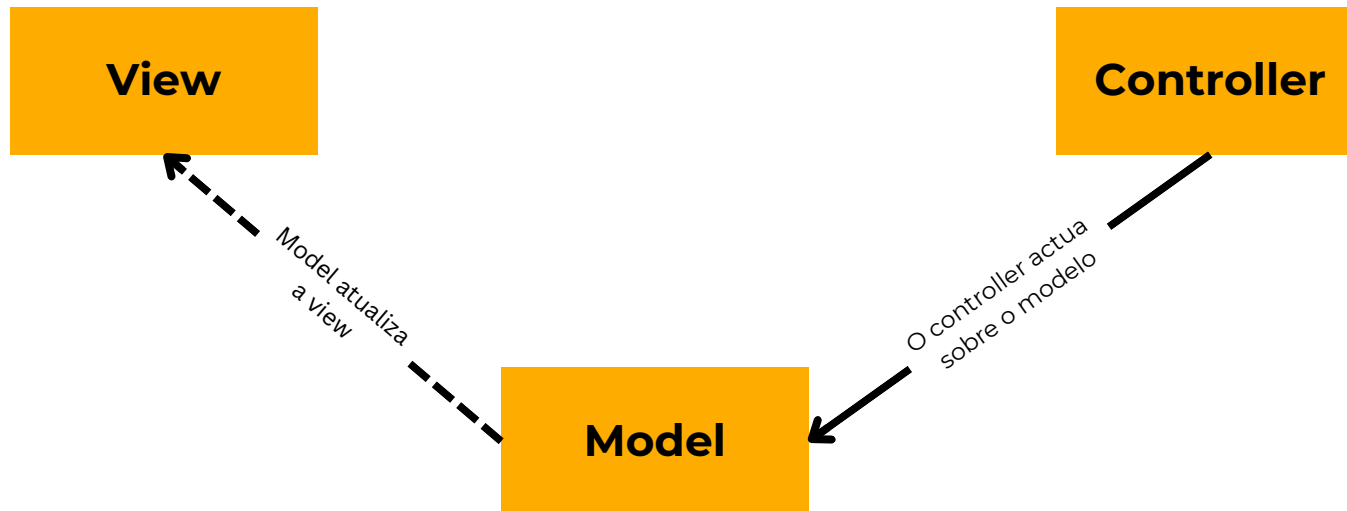


O controller actua sobre o modelo

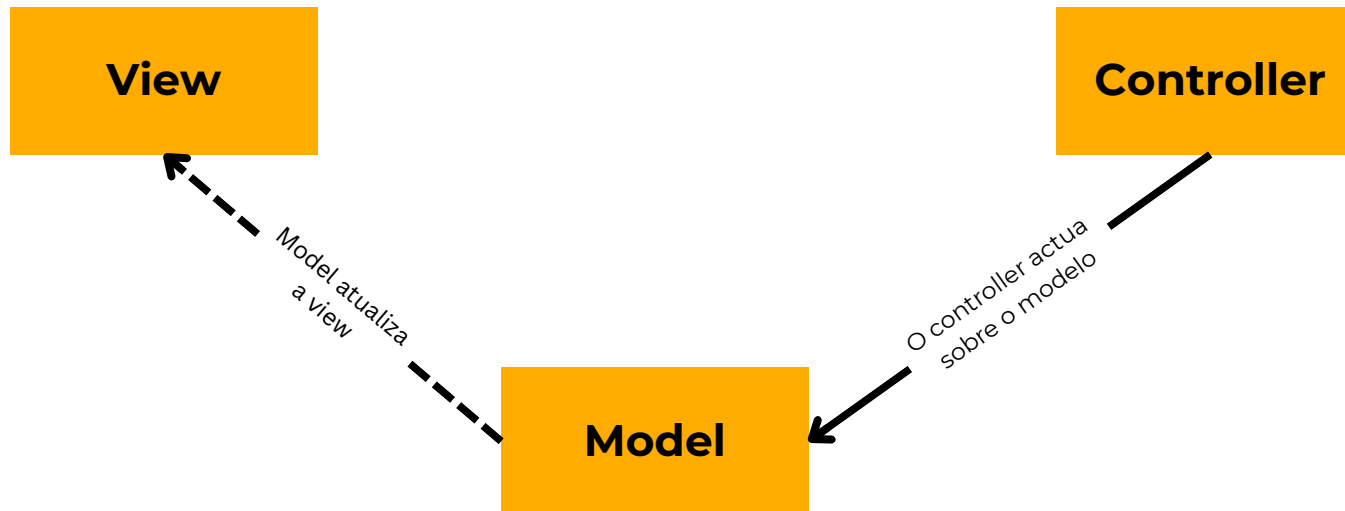
Uma representação (**com estado**) dos dados da aplicação.

Model View Controller

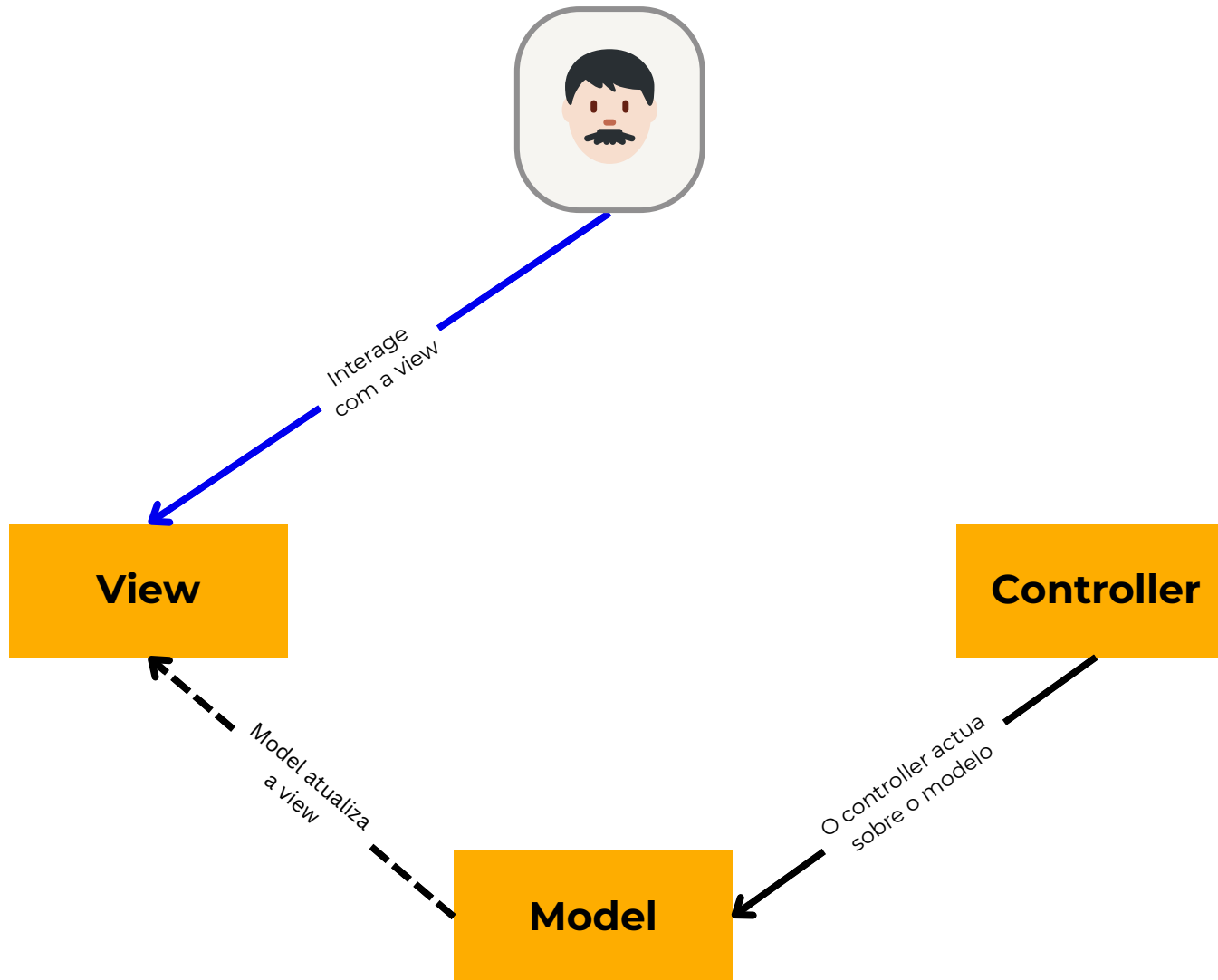
- Cada componente tem uma responsabilidade distinta (*concerns*)
- Ajuda com escalabilidade e manutenção
- Facilita testes ao isolar os componentes.



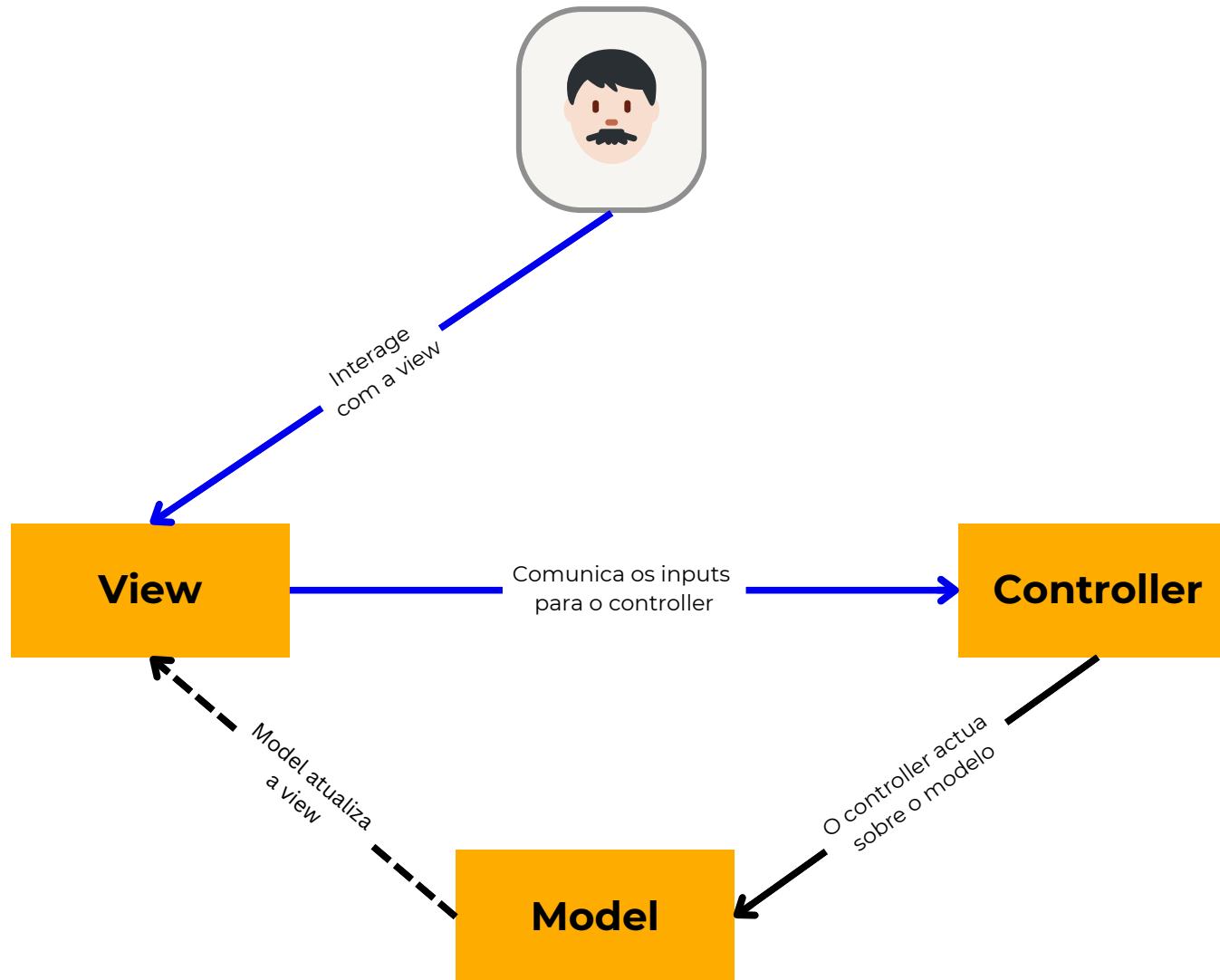
Model View Controller



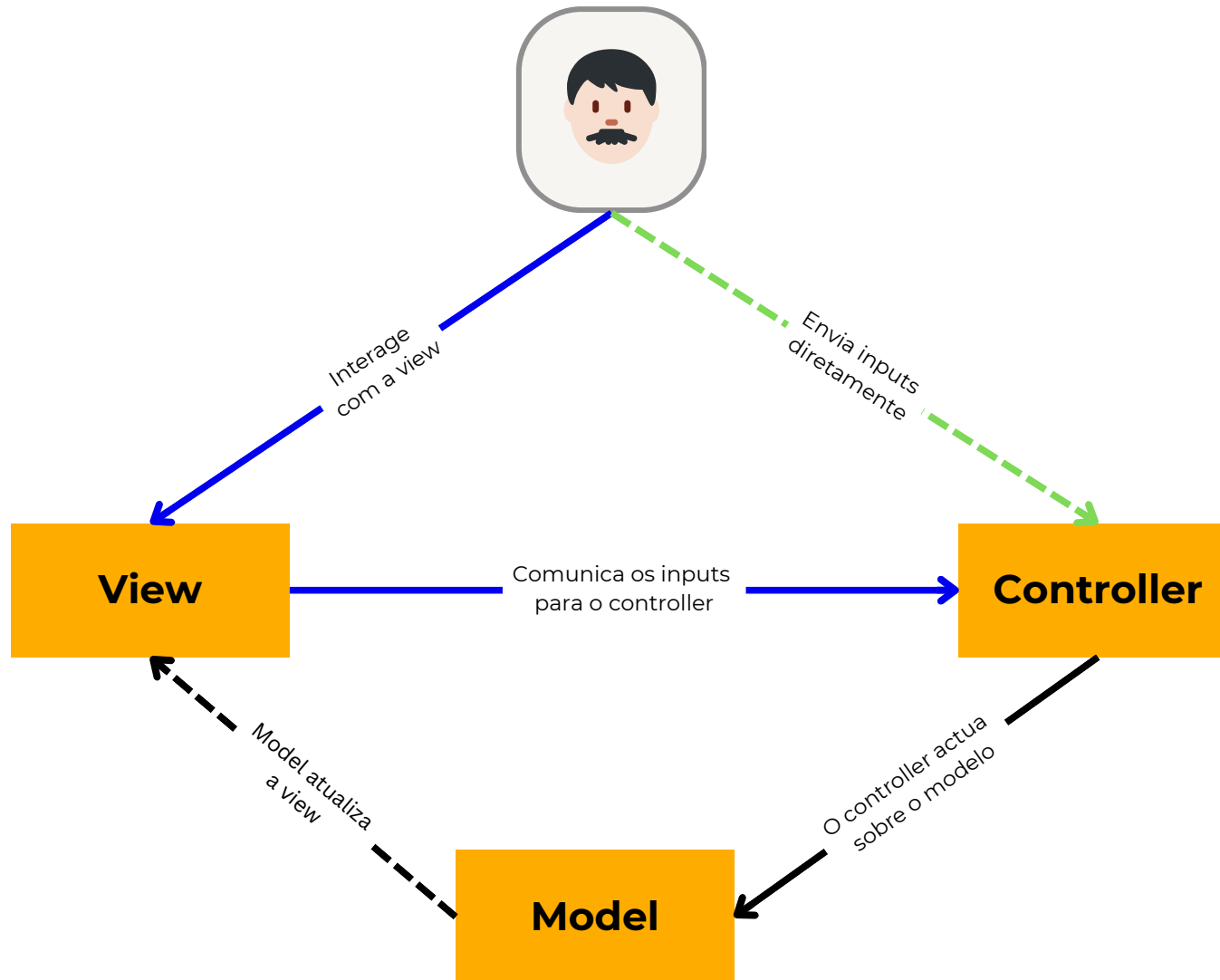
Model View Controller



Model View Controller



Model View Controller



MVC

O bom

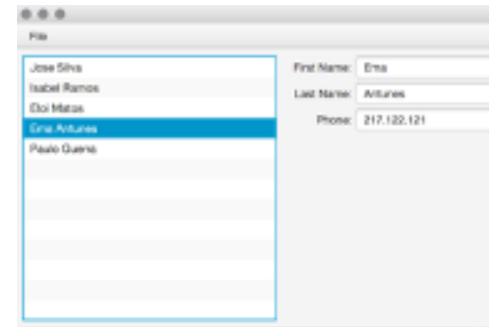
- Separação das responsabilidades bem definida.
- A vista e o modelo estão decoupled:
 - Podem evoluir separadamente (mantendo uma API comum)
 - O mesmo modelo pode ter mais que uma vista possível (ex: contacto na lista, e contacto aberto)
 - Vistas podem ser usadas em vários modelos (ex: lista de contactos, lista de álbuns, etc...), desde que tenham a mesma interface

MVC

O mau

- Manter controlador e vista separados é muito difícil
 - E se a própria vista for modificável pelo controlador?
 - E o estado transiente de interacção, quem o guarda?

- Ex: o elemento que está seleccionado?



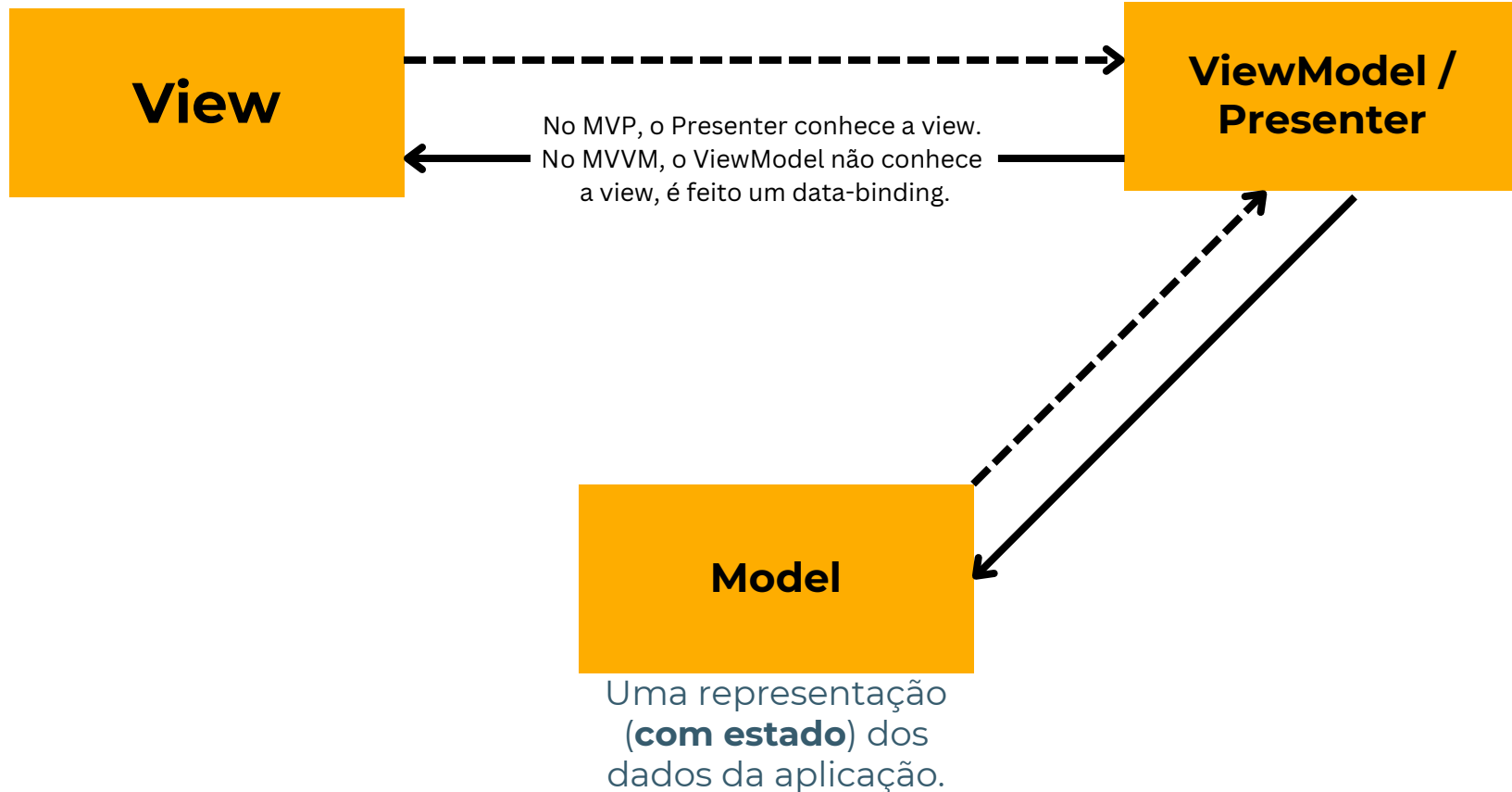
- A comunicação com servidores adiciona latência. O humano quer 0.1s.

MVC

Model-View-Presenter e Model-View-ViewModel

Define a estrutura, aparência dos dados que são mostrados. Mantém o estado dos components e trata do input low-level.

Lida com o input de alto nível, e manipula tanto o modelo como a view. Funciona como intermediário.



Próximas aulas

- Padrões para aplicações nativas (Desktop, Android)
- Padrões para a Web (server-side)
- Padrões para a Web (client-side)

Model-View- Controller

Padrões para a Camada de Apresentação